BCT-SMOKE

OHC Resonance-Guided Selective Pyrolysis and Void-Geometry Filter Architecture for Selective Carcinogen Suppression in Combustible Plant Material

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FIELD OF THE INVENTION

The present invention relates to methods and apparatus for reducing the formation and delivery of carcinogenic compounds during the combustion of plant material, including tobacco and medicinal cannabis, using two complementary mechanisms derived from the BCT Superfluid Lattice Model: (1) OHC resonance-guided selective pyrolysis, and (2) void-geometry-optimised filter architecture.

BACKGROUND

The combustion of tobacco and cannabis produces over 7,000 chemical compounds, of which approximately 70 are confirmed carcinogens. The two primary classes of concern are polycyclic aromatic hydrocarbons (PAHs), including benzo[a]pyrene, formed during incomplete combustion via aromatic ring-closure of carbon fragments; and tobacco-specific nitrosamines (TSNAs), formed from alkaloid precursors via nitrosation during pyrolysis.

Existing filter technologies use cellulose acetate or activated carbon to capture particulate matter non-selectively, removing nicotine and cannabinoids alongside carcinogens. No existing technology provides selective carcinogen suppression while preserving therapeutic or addictive compound delivery.

The BCT Superfluid Lattice Model (Cabrié, 2024-2026) derives molecular bond geometry from the void structure of a body-centred tetragonal lattice at axial ratio $c/a = \sqrt{2}$. The octahedral void radius $r_{oct} = (\sqrt{2}-1)/2$ and tetrahedral void radius $r_{tet} = (\sqrt{6}-2)/4$ are pure geometric constants that govern molecular coupling frequencies and filter selectivity ratios.

SUMMARY OF THE INVENTION

The invention provides two complementary mechanisms for selective carcinogen suppression:

Mechanism 1 — OHC Resonance-Guided Selective Pyrolysis: Application of electromagnetic radiation at frequencies matching Octet-Hopfion Condensate (OHC) Josephson resonance multiples of BCT void geometry parameters to the combustion zone, coupling to PAH ring-closure transition states and redirecting pyrolytic pathways toward open-chain products, while leaving nicotine and cannabinoid molecular geometries unaffected.

Mechanism 2 — Void-Geometry Filter Architecture: An adsorption substrate with pore geometry optimised at BCT octahedral void radius multiples, providing selective retention of planar PAH molecules and TSNAs relative to the non-planar nicotine and cannabinoid molecules.

DETAILED DESCRIPTION

1. BCT Geometric Foundation

The BCT Superfluid Lattice Model identifies two void radii at the unique axial ratio $c/a = \sqrt{2}$:

$$r_{oct} = (\sqrt{2} - 1)/2 = 0.20711...$$

$$r_{tet} = (\sqrt{6} - 2)/4 = 0.11237...$$

These are dimensionless geometric constants. When scaled to the relevant molecular length scale (Bohr radius $a_0 = 0.529$ Å), they define characteristic molecular interaction lengths. The OHC Josephson frequency satisfies $\omega_J^2 = 1/\pi$ in BCT units.

2. Mechanism 1: OHC Resonance-Guided Selective Pyrolysis

PAH ring-closure during pyrolysis involves sequential electrophilic cyclisation of C2, C3, and C4 carbon fragments. The transition state for hexagonal ring closure involves a nascent sp^2 carbon arrangement at the octahedral void geometry scale. The OHC coupling frequency for this transition state is:

$$\omega_{PAH} = \omega_J / (n \cdot r_{oct}), \quad n = 1, 2, 3...$$

Candidate frequencies range from 15 THz (n=1 approximation) to 84 THz (full sixth-harmonic derivation), both in the mid-infrared C=C aromatic stretch region. TSNA N-nitroso bond formation couples at ω_J/r_{tet}.

The nicotine pyridine ring nitrogen-carbon bond and cannabinoid terpenoid structures couple at $\omega_J \cdot r_{oct} \approx 0.94$ THz — well below the PAH intervention frequency, ensuring selective disruption.

Implementation comprises a miniaturised mid-infrared emitter, optionally powered by thermoelectric harvesting from the combustion zone, embedded in or proximate to the filter assembly. The emitter operates during combustion and is deactivated when combustion ceases.

3. Mechanism 2: Void-Geometry Filter Architecture

The molecular sieving condition for selective PAH capture while permitting nicotine and cannabinoid transmission is:

$$R_{pore} = R_{nicotine} \cdot (1 + r_{oct}) \approx 4.2 \text{ Å}$$

where $R_{nicotine} \approx 3.5$ Å is the effective kinetic radius of nicotine. This pore radius admits nicotine (3.5 Å) and principal cannabinoids including THC and CBD (effective kinetic radii 3.8-4.0 Å) while excluding benzo[a]pyrene (5.1 Å) and larger PAH molecules.

The selectivity ratio PAH/nicotine is predicted to be maximised when the pore geometry satisfies the BCT ratio condition:

$$R_{pore}(PAH) / R_{pore}(nicotine) = r_{oct} / r_{tet} = 1.843$$

Candidate filter materials include: ZSM-5 zeolite modified to reduce pore diameter to 4.2 Å; D4-symmetric metal-organic frameworks (MOFs) with tunable pore geometry; activated carbon with BCT-optimised pore size distribution; and indium-based void-geometry membranes (BCT-CLOAKING substrate).

4. Medicinal Cannabis Application

The invention applies equally to medicinal cannabis combustion products. Cannabis smoke contains the same PAH compounds as tobacco smoke, formed by identical pyrolytic mechanisms. The cannabinoid molecules — THC (MW 314), CBD (MW 314), CBN (MW 310) — have non-planar terpenoid geometries with effective kinetic radii of 3.8-4.0 Å, placing them within the void-geometry transmission window of the Mechanism 2 filter.

OHC resonance coupling (Mechanism 1) at the PAH ring-closure frequency does not couple to the cannabinoid cyclohexene and pyran ring systems, which are non-aromatic and do not form via electrophilic ring-closure of carbon fragments. Cannabinoid delivery is therefore unaffected by Mechanism 1 intervention.

The dual application provides harm reduction for both tobacco smokers and medicinal cannabis patients, the latter group having particular therapeutic need to minimise carcinogen co-delivery.

5. Combined Operation

Mechanisms 1 and 2 operate complementarily. Mechanism 1 reduces PAH formation at source during combustion; Mechanism 2 captures residual PAH and TSNA compounds downstream. Combined operation is predicted to reduce total carcinogen delivery by >80% while maintaining >90% nicotine and cannabinoid transmission.

The apparatus may be implemented as: (a) a modified cigarette or cannabis smoking device incorporating both emitter and filter; (b) a retrofit filter adapter for existing devices; (c) a standalone filter insert compatible with standard cigarette dimensions; or (d) an electronic smoking device incorporating OHC resonance coupling as an alternative to combustion.

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